Permutation Test Analysis: Florida Temperature Trends

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Introduction

This analysis investigates whether there is a statistically significant correlation between year and temperature in the Key West, Florida dataset spanning 1901-2000. A permutation test was conducted to assess the significance of the observed correlation coefficient.

Methods

- Dataset: Key West Annual Mean Temperature (1901-2000)
- Observed correlation between Year and Temperature: $r = 0.533$
- Permutation approach: 10,000 random shuffles of temperature values
- Test statistic: Pearson correlation coefficient
- Null hypothesis: No correlation between year and temperature
- Alternative hypothesis: Non-zero correlation (two-sided test)

Results

Statistic	**Value**
Observed Correlation	0.533
Mean Permuted Correlation	-0.001
Standard Deviation	0.101
p-value	0.000

Interpretation

The permutation test reveals strong evidence against the null hypothesis:

- The observed correlation of 0.533 indicates a moderate positive relationship between year and temperature
- None of the 10,000 permuted datasets produced a correlation coefficient as extreme as the observed value
- The resulting p-value of approximately 0.000 provides strong statistical evidence ($p < 0.001$) that the correlation is not due to random chance

This suggests a statistically significant warming trend in Key West, Florida over the 20th century, consistent with broader climate change patterns observed in many regions.

permutation_histogram.pdf

Figure 1: Distribution of 10,000 permutation correlation coefficients. The red line indicates the observed correlation (0.533).